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Labs: Writing Papers
Image Based Biometry

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Content

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Writing sections
Overview

- What is the IMRAD structure?
- In what order do we write?
- What goes where?
- How to write each part.
- Overview of some actual examples.

Introduction

- The content of these slides is based on personal experiences from reviewers' comments and the knowledge of more experienced colleagues. They are still only opinions.
- However, these guidelines are helpful in all technical writing, not only scientific papers.

IMRAD

- The structure of IMRAD:
- `https://en.wikipedia.org/wiki/IMRAD`
-
  - **I**ntroduction
  - Related Work
  - **M**ethod
  - Experiments, **R**esults **and** **D**iscussion
  - Conclusion
- What is the order of writing?

IMRAD

The recommended order of writing:

- Do research on related work and make notes (and finalize it whenever),
- Start working on experiments, make notes and as soon you get good results, continue with the paper,
- Methodology,
- Experiments and Results,
- Discussion and Conclusion,
- Introduction,
- Abstract.

Feel free to write Experiments or Results before Methodology, but definitely achieve good results before writing up the paper and definitely finish with Abstract.

Tools

- `https://notebooklm.google.com/`
- LaTeX (recommended online compiler: `http://www.overleaf.com`)?
  - WYSWYG tools (e.g. Word/Writer) are great for writing something shorter, but they are not really made for writing papers: no word breaking, referencing more difficult, equations more difficult, table of contents more difficult, version incompatibility, everything looks worse.
  - It is not all perfect: `https://tex.stackexchange.com/questions/1756/why-should-i-use-latex`.
  - The negative aspect of difficulties compiling Tex can be removed by using online compilers.

General Tips

Writing:

- Avoid using too complicated terms.
- Avoid too complicated sentence structures or very long sentences.
- Avoid adverbs (very, etc.).
- When using an acronym make sure to explain it the first time you are using it.
- Prepare diagrams to support the explanation of your methods.
- Imagine that the reader will be the laziest person on Earth.

General Tips

- Be careful not to repeat yourself.
- The use of active/passive voice is a delicate topic. Generally stick to more active voice; it reads better and is easier to understand for the reader.
- Which is better:
- *"We use a histogram equalization to improve recognition performance."*
- *"During recognition, to improve performance, a histogram equalization is performed."*

Using Present/Past Tense

Using Present/Past Tense:

- Past tense for the related work, what you did and when describing results.
  E.g.: "Xiang showed that, ...", "We found that, ...", "We proposed smth, ...".
- Present tense for for covering conclusions/indications and for future work.
  E.g.: "The results show that, ...".

General Tips

- Make sure you reference every single figure/table/equation.
- Make figures clear, font sizes should match.
- Plots should be in vector format.
- Make images of the correct size and make them visible.
- Check for "Figure X" vs "Fig. X".
- Captions should stand on their own.
- Do not use too many borders in tables, instead make use of white space (if possible).

Writing Abstract

Abstract:

- Despite being one short paragraph, it consists of three parts:
  - Motivation: What is the current status of the field? What is the problem?
  - Approach: How did you solve the problem?
  - Findings: What are the results and implications?
- Stands on its own, this means:
  - no references,
  - no acronyms (some exceptions),
  - one paragraph.
- Draft/generate multiple versions and then polish, polish, polish. Often the only thing people read.

Why does the abstract need to stand on its own?

Writing Introduction

- Roughly consists of the following:
  - State: Current status of the field (for better results provide LLM with your own list of important papers).
  - Problem: Presentation of the problem (what is currently missing, what is not researched yet, problematic etc.).
  - Solution: Brief description of your solution to the problem, how the solution is structured.
  - Paper structure (optional): Briefly describe what is where in the paper.
- Include an overview diagram of your method already in the introduction. Use generative AI to get diagram drafts

Writing Related Work

- Only after you have a list of papers, use LLMs to compile the text.
- Adapt to the target audience (e.g. field of the journal).
- Avoid describing into too much details, rather provide a reference and instead cover more work. Make sure that you describe the core idea and not trivial parts of that work.
- Try to cite the initial, original work where applicable.
- Including a tabular overview is especially useful, but not that common.

Writing Methodology

Methods:

- Arguably, the most important part of the paper.
- Be clear, include diagrams.
- Within each paragraph try to go from a general to the specific idea, whilst retaining one idea per paragraph.
- Title the section appropriately.
- No details about the experiments.
- Make sure to describe your approach as well as possible when feeding it into LLMs.

Writing Experiments, Results & Discussion

Results & Discussion:

- Describe experiments well and present results clearly. Your guide should be, that the reader is able to reproduce your results easily.
- Figures are for displaying a larger overview, whereas tables are for the exact values. Do we need both?
- Discuss not only through the numerical data, but also generalize, what are the trends, possible conclusions, why do you think something is, etc.
- Nowadays when writing text is cheap and quick, you will spend the most time preparing this (nice plots, good tables).

Writing Conclusion

Conclusion roughly consists of three parts:

- Summarize what you have done.
- Describe the findings through advantages and also mention limitations. This is good, not only for the community, but also you are ready for the reviewers.
- Wrap it up with what the larger implications are and what are the possible future improvements or where do you think the research will go.
- If your whole paper fits into the LLMs' context your golden. Generate multiple versions and polish.

Preparing References

References:

- Be careful with LLMs here, as mostly they still produce crap and make works (authors, titles, URLs) up.
- Make sure you do not miss the latest state-of-the-art.
- Do not include papers that are not important to the work presented in the paper.
- Make sure you do not miss any of the important work that was done. Not covering work from e.g. the last two years can mean two things (both are bad):
  - you did not check the literature well enough, meaning your approach could be obsolete,
  - or actually nothing happened in the last two years, meaning the research has shifted and the whole area is obsolete.
- Make sure that styles match, i.e.: bibtex from Scholar or from IEEEXplore differ, one will contain DOI, other will not, or one will have conference name written differently than the other.

Example #1



Figure 1. Ear images

DSIFT features result in a lower performance, while the binary-pattern-based techniques again perform better and more or less on the same level [**?**]

When we focus only on the LBP we can say that this method may have some difficulties in identifying some characteristics because we have some ears that have some "obstacles" like earrings, hearing aids and in some cases where the colour of the ear is very similar to the colour in the background of the image. This method will also have difficulties if the texture of the image is not in the range that is recognized by LBP is a particular case of the texture spectrum.

Missing citation references and maybe not the best way to arrange images. To solve errors like this, search for "?" before you submit your paper.

Example #2

duce a large variation when perturbed
ıer detection algorithm is a mathematical
ing such differences in which we may even
as eigenvalues to determine how much of
in a given window. As the Harris corner
is largely invariant to rotation and scale,
to try and find all high-frequency corners
nine their relative positions. The idea is
v we take the picture of the same ear, it
corners at similar intervals to other ears
me person in our database. Harris Corner
included in the native MatLab library.

Net is the name of a specific architecture of
ːal network designed to classify hundreds of
was trained on 1.2 million high-resolution
ıng the top-1 error rate of about 37.5%. One
ibutors to its success was a large database,
usion of a regularisation technique called
ːdomly disables some portion of neurons in
yer in order to prevent overfitting. As the
ork is a class and we are only interested
ld be used to describe our ears, we extract

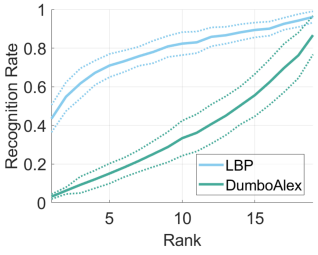


Figure 3. A CMC curve comparison of DumboAlex with LBP

The probe image is tested against the *rank*-nearest neighbours of every image in our dataset. By the CMC (Cumulative

Font sizes do not match.

Example #3

***Abstract*—In this paper we will present experimental evaluation of local phase quantization and dense scale invariant feature transform methods on AWE dataset.**

I. Introduction

Automatic identity recognition from ears has been researched increasingly in the last decade. As fingerprint and iris recognition are well researched some other methods for identification still offer a lot of potential for application.



Figure 1. This is example of an image that we predicted that classification will fail due to the occlusion and accessories.

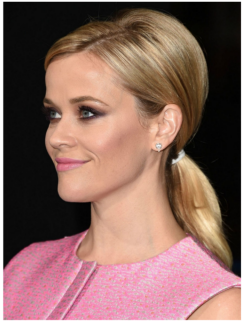


Figure 2. This is example of an image that we predicted that classification will succeed because ear is fully visible with good lightning.

B. Dense Scale Invariant Feature Transform

SIFT descriptors are invariant to scaling, translation and rotation. This descriptors are calculated on a local neighbour-

Abstract too short, images too large.

Example #4

References

1. *Histogram of oriented gradients*, Feb 2018.
2. Ziga Emersic, Vitomir Struc, and Peter Peer, *Ear recognition: More than a survey*Ear recognition: More than a survey, CoRR **abs/1611.06203** (2016).
3. *Local binary patterns*, Feb 2018.

Are these references? :)

Example #5

III. Results

A.HOG

In parameters tuning for HOG I conducted experiments on the several parameters sets.

1.Parameter set: cell size[32 32]px,block size [2 2],orientation 9,block overlapping 50%

2.Parameter set: cell size[32 32]px,block size [2 2],orientation 18,block overlapping 50%

3.Parameter set: cell size[32 32]px,block size [2 2],orientation 9,block overlapping 0%

4.Parameter set: cell size[16 16]px,block size [2 2],orientation 18,block overlapping 50%

5.Parameter set: cell size[16 16]px,block size [2 2],orientation 9,block overlapping 50%

6.Parameter set: cell size[8 8]px,block size [2 2],orientation 18,block overlapping 50%

Not very readable, is it?

Example #6

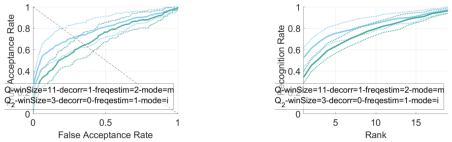


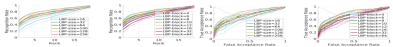
Figure 4. ROC (left) and CMC (right) curve of LPQ method.

proper annotations for the dataset.

For feature extraction it is crucial for good performance of the method to choose the right, optimal, parameters. We measured the performance using the AUC score, ROC and CMC curve. Interpretation of which gave us optimal parameters

Plot labels not readable, fonts also not consistent compared to the text.

Example #7



(a) Input image size (b) Block size (c) Neighbourhood radius (d) Number of bins

Figure 2: CMC Curves for LBP descriptor

(a) Input image size (b) Block size (c) Neighbourhood radius (d) Number of bins

Figure 3: ROC Curves for LBP descriptor

D. Testing

All extractors we test with AWE toolbox and dataset. In AWE dataset are 1000 images of 100 subjects. We test extractors with 5-cross-validation, which is predefined in AWE toolbox. In extractors, we first test every specific parameter and observe how it perform. Then we take some best and default values for every parameter and try different combinations. So we get the best parameter for the specific extractor. On end, we try to improve our results with sharpening image before extractions. For evaluating extractors and their parameters we use Cumulative Match Curve (CMC) for identification problem and Receiver Operating Characteristic (ROC) curve for verification problem.

III. Results

In this section, we represent the results of our evaluation. We try to test different parameters. Boundaries of parameters we decide based on if we increase parameter than performance must improve. If it not improve then we stop testing. We stop too if runtime of evaluating parameter is to slow.

First, we analyze LBP extractor. We can see results on images[2]and [3]. For input image size we can see that performance improve for CMC and ROC curve and is best for size 64,100, 128. After that, it drops. Very similar happen with radius (samples) parameter. Here we get the best score at radius 6 to 10. For block size, we see that bigger than block is, worse performance is in our extractor. This probably happens because with a smaller size of the block we can compute more different descriptors and because of that we are more accurate. For a number of bins, we see that when we have more than 8 bins we get always a very similar result. It is very interesting that we get best results in CMC and ROC curve for same parameters. After our test with different parameter combinations, we decide that best parameters for LBP are size=64, block=4, bins=58, samples=8 (Image[4]).

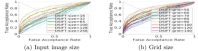
Results of our evaluation for DSIFT descriptor are represented on images[4] and [5]. Here we can see that descriptor perform best on input image size of 100 and 128, grid size 90 and 100, patch size between 4 and 16, number of bins between 8 and 32. Same as for LBP we get here again best result for CMC and ROC for the same parameters. After combination test best parameters for DSIFT extractor are input size of image=128, patch size=8, grid=100, bins=8 (Image[6]).

As we can see from Image[6] we improve our default result of extractors. On end, we try with a sharpening of the image.



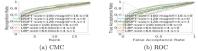
(a) Input image size (b) Grid size (c) Patch size (d) Number of bins

Figure 4: CMC Curves for DSIFT descriptor



(a) Input image size (b) Grid size (c) Patch size (d) Number of bins

Figure 5: ROC Curves for DSIFT descriptor



(a) CMC (b) ROC

Figure 6: Best results for tested extractors and result with sharpening of image (s=1)

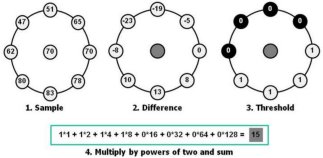
While the author tried to present as many results as possible, these plots are just not readable.

Example #8

Metoda lokalnih binarnih vzorcev (ang. local binary patterns - LBP) je preprosta a učinkovita metoda, ki označi piksle na podlagi upragovanja njihove okolice (slika 2). Vsaka oznaka je predstavljena v obliki binarnega števila. Histogram teh števil lahko uporabimo kot vektor značilk [lbp].

The value of the LBP code of a pixel (x_c, y_c) is given by:

$$LBP_{P,R} = \sum_{p=0}^{P-1} s(g_p - g_c)2^p \qquad s(x) = \begin{cases} 1, if\ x \geq 0; \\ 0, otherwise. \end{cases}$$



Slika 2: Izračun oznake piksla z metodo LBP.

Do not use diagrams/figures from other people, especially while not citing their work. Furthermore, the screenshot of the figure is of bad quality (e.g. gray background).

Example #9


```
********RESULTS********
Rank-1           : 30.84+-5.65 (10)
EER_er           : 35.11+-8.80 (1775)
VER_01FAR_ver       : 4.18+-2.88 (1470)
VER_1FAR_ver        : 14.87+-6.03 (1535)
AUC          : 68.74+-8.42 (3525)
#min(FSc/n)      : 1026
#min(TSc/n)      : 55
#min(Sc/n)       : 1083
#Sc      : 12062
**********************
********RESULTS********
Rank-1           : 41.84+-8.31 (10)
EER_er           : 32.95+-8.15 (2046)
VER_01FAR_ver       : 4.57+-4.07 (1291)
VER_1FAR_ver        : 16.49+-5.59 (1499)
AUC          : 73.65+-8.88 (3712)
#min(FSc/n)      : 1026
#min(TSc/n)      : 55
#min(Sc/n)       : 1083
#Sc      : 12062
**********************
```

Figure 1. The results from implementing HOG with (32,32) an (8,8) cell size

Someone was in a hurry! :) Solution: start writing the report/paper earlier.